ASSIGNMENT-2

Foundations of Cryptography

Q1. Four Fundamental Cryptographic Goals

The four fundamental goals of cryptography are:

• **Confidentiality:** Ensures that only authorized people can read the information.
• **Integrity:** Ensures that information is not changed, deleted, or damaged without authorization.
• **Authentication:** Helps verify the identity of a person, device, or system involved in communication.
• **Non-repudiation:** Provides evidence that a particular person or system performed an action or sent a message and makes it difficult for the sender to deny it later.

Together, these objectives help create secure and trustworthy communication between users and computer systems.

Q2. Confidentiality

Confidentiality means protecting information from unauthorized access or disclosure. Only the intended or authorized users should be able to view the information.

Example: When a customer enters a password or card information on a secure banking website, encryption helps prevent unauthorized people from reading that information while it is being transmitted. Thus, confidentiality keeps sensitive data private and protects it from being exposed to attackers.

Q3. Four Cryptographic Objectives in Detail

Cryptography provides four important security objectives that work together to protect communication.

1. Confidentiality
Confidentiality ensures that only authorized users can read information. Encryption is commonly used for this purpose. **Example:** Online banking encrypts sensitive information during communication.

2. Integrity
Integrity ensures that data is not changed without authorization. Hash functions and message authentication mechanisms can help detect modifications. **Example:** A downloaded file can be checked to see whether it was changed during transfer.

3. Authentication
Authentication verifies the identity of communicating parties. Passwords, digital certificates, and digital signatures can support authentication. **Example:** A browser verifies a website using its digital certificate.

4. Non-repudiation
Non-repudiation provides evidence of who created or signed information. Digital signatures are commonly used. **Example:** A digitally signed document can show that it was signed by the holder of a particular private key.

Why all four are important: Confidentiality alone does not prove that data was not changed or identify the sender. A secure system therefore needs a suitable combination of confidentiality, integrity, authentication,

and non-repudiation.

Q4. Substitution Cipher vs Transposition Cipher

A **substitution cipher** replaces each plaintext symbol with another symbol according to a rule or key. The positions of the symbols generally remain the same. **Example:** In the Caesar cipher, each letter is shifted by a fixed number of positions, such as A becoming D when the shift is 3.

A **transposition cipher** keeps the original letters but changes their positions according to a particular pattern. **Example:** The Rail Fence cipher writes a message in a zigzag pattern and then reads the rows in sequence.

Main difference:
• Substitution changes the **symbols**.
• Transposition changes the **positions** of symbols.

Both are classical cryptographic techniques, but modern cryptography uses much stronger mathematical methods.

Q5. Evolution from Classical Ciphers to Modern Standards

Cryptography developed from simple classical methods to strong computer-based algorithms because older techniques became easier to break as cryptanalysis and computing power improved.

Classical Ciphers: Early methods such as the Caesar and Vigenère ciphers mainly used substitution. They were useful but vulnerable to frequency analysis and other attacks.

DES: The Data Encryption Standard became an important computer-based encryption standard. However, its effective 56-bit key eventually became too short because computers became powerful enough to perform large brute-force searches.

AES: AES was introduced as a stronger replacement for DES. It supports 128-, 192-, and 256-bit keys and is widely used today because of its strong security and good performance.

RSA: RSA introduced an important public-key approach for encryption and digital signatures and helped address key-distribution difficulties.

SHA: SHA refers to a family of cryptographic hash algorithms used for integrity checking and other security applications. Stronger versions were developed as weaknesses were discovered in older hash functions.

The main motivation for these changes was the need for stronger security against improving computers, cryptanalysis, and new forms of attack.

Q6. Key and Key Management

A **key** is a value used by a cryptographic algorithm to control encryption, decryption, authentication, or other security operations. The key works together with the algorithm to protect information.

Key management means securely creating, distributing, storing, using, rotating, and destroying

cryptographic keys. It is important because even a strong cryptographic algorithm can become insecure if its key is stolen, guessed, incorrectly shared, or poorly stored.

For example, in symmetric encryption, the communicating parties must securely share the secret key. In asymmetric cryptography, the private key must be strongly protected because anyone who obtains it may be able to perform actions as its owner. Good key management is therefore an essential part of a complete security system.

Q7. Cryptography in Computer Network Security

Cryptography plays an important role in protecting computer networks and the information transmitted through them.

1. Protecting Data During Transmission: Encryption converts readable information into ciphertext so attackers cannot easily understand it. **Example:** HTTPS uses cryptographic techniques to protect communication between a browser and a website.

2. Preventing Unauthorized Access: Authentication mechanisms such as passwords, certificates, and cryptographic keys help verify users and systems. Access can then be limited to authorized parties.

3. Detecting Tampering: Cryptographic hash functions and message authentication mechanisms can help detect whether information has been changed. If the received data produces a different expected value, modification may have occurred.

4. Supporting Secure Communication: Digital signatures can provide authentication and integrity, while key-exchange techniques help establish secure session keys.

Therefore, cryptography helps networks protect privacy, verify identities, maintain data integrity, and reduce the risk of unauthorized modification.

Q8. Passive Attack vs Active Attack

A **passive attack** occurs when an attacker observes or monitors information without changing the data or communication. The main aim is usually to obtain information secretly.

Example: An attacker secretly monitoring network traffic to learn sensitive information is a passive attack.

An **active attack** occurs when an attacker changes, inserts, deletes, or disrupts information or communication. The attacker actively interacts with the system.

Example: An attacker modifying a message while it is being transmitted is an active attack.

Difference:
• Passive attack = **observes** information.
• Active attack = **changes or interferes** with information.

Encryption can help protect against information disclosure, while integrity checks, authentication, and other security controls help detect or prevent many active attacks.

Q9. Historical Evolution to Counter Security Threats

Cryptographic techniques have evolved because attackers and computing technology have also evolved. Early classical ciphers, such as Caesar and Vigenère, provided basic protection but could be attacked using mathematical and statistical analysis.

With computers, stronger symmetric algorithms were developed. DES became an important standard, but its 56-bit key eventually became vulnerable to brute-force attacks as computing power increased. AES was later developed as a stronger and more efficient modern standard.

A major change occurred with the introduction of **asymmetric or public-key cryptography**. Traditional symmetric systems had a major key-distribution problem: two parties needed to securely share the same secret key before communication could be protected.

In 1976, **Diffie and Hellman** introduced the public-key concept and Diffie-Hellman key exchange, showing how parties could establish a shared secret without directly transmitting that secret. Later, **RSA** provided another important public-key system supporting encryption and digital signatures.

Thus, cryptography moved from simple manual methods to stronger computer-based systems designed to handle larger networks and increasingly powerful attacks.

Q10. Modern Symmetric and Asymmetric Algorithms

One modern symmetric algorithm is **AES (Advanced Encryption Standard)**. AES uses the same secret key for encryption and decryption and is widely used to protect stored and transmitted data. It supports 128-bit, 192-bit, and 256-bit keys.

One modern asymmetric algorithm is **RSA**. RSA uses a public key and a private key and can be used for public-key cryptographic operations such as encryption and digital signatures.

In modern systems, symmetric and asymmetric algorithms are often used together. Asymmetric cryptography can help with authentication and establishing or protecting keys, while a fast symmetric algorithm such as AES can encrypt the actual large amount of application data.

Q11. Classical and Modern Cryptography

Cryptography can broadly be divided into **Classical Cryptography** and **Modern Cryptography**.

Classical Cryptography: Classical methods were often performed manually or with mechanical devices. They included substitution and transposition techniques. Examples include the **Caesar Cipher** and **Vigenère Cipher**. The Enigma machine during World War II was an important mechanical development. These systems were limited compared with modern computer-based cryptography and could be attacked using cryptanalysis.

Modern Cryptography: Modern cryptography uses computers, mathematics, algorithms, and carefully designed protocols. **DES** became an important early computer-based standard, while **AES** provided stronger modern symmetric encryption. Public-key cryptography introduced systems such as **RSA**, which use public and private keys. Cryptographic hash functions such as **SHA** are used for integrity and other security purposes.

The transition happened because digital communication required stronger security, automation, scalability, and protection against increasingly powerful computers and attackers.